Mitchell et al. (1982) solution

Mitchell et al. (1982) conducted physical model tests on cemented backfill to determine the required strength of stable backfill when the vertical fill face is exposed. They considered a 3D geometry of cemented backfill block with an open front face, as shown in Fig. 5.15. In this figure, L, B, and H are the backfill length (perpendicular to strike), width (parallel to strike) and height respectively. A sliding plane inclined at an angle ($\alpha = 45^\circ + \frac{\phi}{2}$) to the horizontal is presumed. Equations 5.41, 5.42, and 5.43 are the developed solutions for the factor of safety (FoS), required cohesion (c), and UCS, respectively.

The factor of safety (FoS) is given by,

$$\mathrm{FoS} = \frac{W_n \cos\alpha \tan\phi + cLB/\cos\alpha}{W_n \sin\alpha} = \frac{\tan\phi}{\tan\alpha} + \frac{2cL}{H^* (\gamma L - 2C_s) \sin 2\alpha} \quad \text{......... (5.41)}$$

where, ϕ is the angle of internal friction of backfill on the sliding plane, c is the apparent cohesion (kPa) of backfill on the failure plane, α is taken to be $45^\circ + \frac{\phi}{2}$ (critical plane), γ is the backfill unit weight (kN/m^3), C_s is the cement bond shear strength (kPa), $W_n = BH^*(\gamma L - 2C_s)$ is the net weight of sliding block and $H^* = H - (B \tan\alpha)/2$.

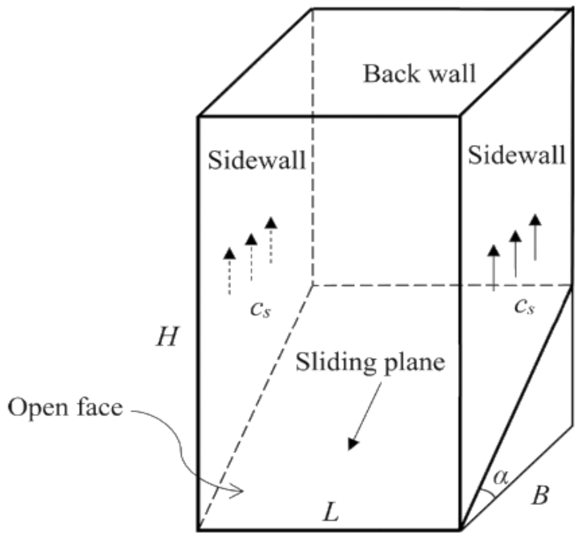

Fig. 5.15: Sliding wedge block model of Mitchell et al. (1982) solution.

The required cohesion (c) is given as,

$$c = \gamma H/2(\frac{H}{L} + \tan\alpha) \quad \text{.........(5.42)}$$

The required UCS can be determined by using equation 5.43,

$$\mathrm{UCS} = \gamma H/(1 + H/L) \quad \text{.........(5.43)}$$

These above solutions were developed considering the following assumptions:

- The wedge block is under shear stresses along the interfaces between the backfill, foot wall (F/W) and hang wall (H/W). The shear stresses are due to bond cohesion only, where $C_s = c$, backfill-rock wall interface friction angle (δ) = 0.
- The stresses (normal and shear) on the stope back wall (opposite of the exposed fill face) is assumed zero.
- The stope is narrow and high, i.e., $H \geq B\tan\alpha$.
- There is no surcharge loading on top of the backfill.
- Equation 5.43 assumes a frictionless backfill (ϕ = 0), and UCS = 2c.
- The required cohesion of backfill (Eq. 5.42) is derived considering the factor of safety (FoS = 1).

The FoS value depends upon various parameters, such as the level of control implemented on backfill mix, backfill slurry quality control, ground conditions, stope excavation methodology, height of exposed stope and pillar recovery methodology. From the points of view of the degree of uncertainties and scale of mining operations, the statutory regulating body, Directorate General of Mines Safety (DGMS), India, suggest FoS of backfilled stope to be 1.5 to 2 depending on the ground conditions for safer mining practices in India. Singh et al. (2019) also used FoS of 2 for the required strength estimation for a backfilled stope. Based on the requirements in Indian scenario, the equation developed by Mitchell et al. (1982) is modified with FoS of 2 for determination of required cohesion (c) of backfill and can be determined by using equation 5.44.

$$c = \gamma H/(\frac{2H}{L} + A) \qquad \text{.........(5.44)}$$

where, $A = \frac{1}{(\sin 2\alpha - \cos^2 \alpha . \tan \phi)}$ ....(5.45)

Similarly, Arioglu (1984) modified the equation developed by Mitchell et al. (1982) by using $H/B \geq \tan\alpha$ as 3.15 MPa. Chen and Jiao (1991) also worked on the studies carried out by Mitchell et al. (1982) for estimating the required strength by using FoS of 3 to 4 for Canadian mine having cemented sand filling.

For assessment of safe and critical height of the backfill, Zou and Nadarajah, 2006 worked on the empirical equation developed by Mitchell et al, 1982 by varying the backfill cohesion and considering a surcharge load on top of the backfill. Dirige et al. (2009) also, for determination of stability of backfill in inclined stopes with rough rock wall condition modified the empirical equation as per the site condition. Considering the limitation of high aspect ratio (HAR), so that the sliding plane does not touch the top of the backfill and remain within the fill i.e., H/B ≥ tan α, modifications were further made by Li and Aubertin (2012).

5.8.3 Li and Aubertin (2012) solution

Li and Aubertin (2012) adopted the original Mitchell et al. (1982) solution by considering the following in addition to the basic assumptions of the original solution:

- The likelihood of the presence of surcharge load P_o (kPa) on top of the backfill (Fig. 5.16) due to heavy equipment, additional backfill, and broken rock material on top of the backfilled stope.
- A non-zero backfill friction angle ($\phi > 0$), different backfill-wall rock interface cohesion (c_s) and backfill cohesion (c). This distinction of cohesion has been termed with a ratio named as adherence ratio ($r_s = 0$ to 1), $c_s = r_s c$.
- Different aspect ratios, i.e., high aspect ratio (HAR): H/B ≥ tan α and low aspect ratio (LAR): H/B < tan α.

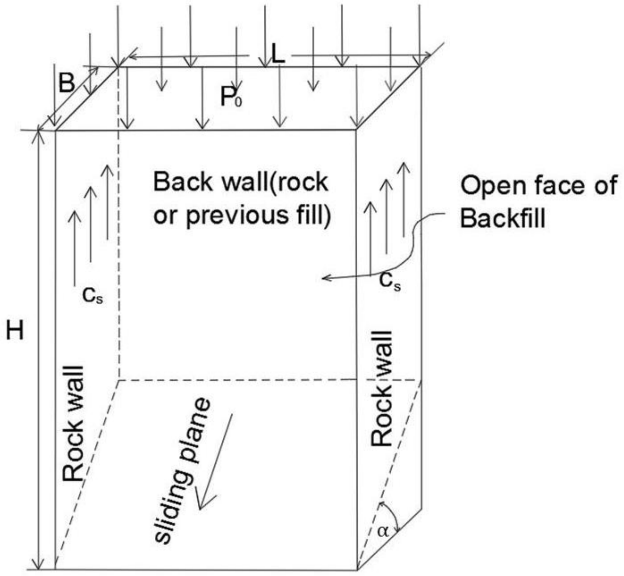

Fig. 5.16: Wedge block model with overlying load (P_0) on top of backfill (Li & Aubertin, 2012).

Considering the case of a HAR stope, the solution for FoS can be determined by using equation Eq. 5.46.

$$\mathrm{FoS} = \frac{\tan\phi}{\tan\alpha} + \frac{2c}{\left[P_o + H^*\left(\gamma - \frac{2c_s}{L}\right)\right]\sin 2\alpha} \qquad \text{………(5.46)}$$

The required cohesion for a stable backfill is given by Eq. 5.47.

$$c = \frac{\frac{(P_o + \gamma H^*)}{2}}{\left[\left(FS - \frac{\tan\phi}{\tan\alpha}\right)\sin(2\alpha)\right]^{-1} + \frac{r_s H^*}{L}} \qquad \text{………(5.47)}$$

$$H^* = H - (w \tan\alpha)/2$$

Here, w is the same that of B, Considering the failure of backfill to follow the Coulomb criterion, the expression for required UCS can be expressed in terms of c and ϕ and can be determined by using equation 5.48.

$\mathrm{UCS} = 2c\tan(45° + \phi/2)$ ………(5.48)

However, for the cases of LAR, the potential sliding plane will intersect the backfill top. The FoS and the required backfill cohesion for such cases can be determined by using equations 5.49 and 5.50 respectively.

$$\mathrm{FS} = \frac{\tan\phi}{\tan\alpha} + \frac{2c}{\left[P_0 + H\left(\frac{\gamma}{2} - c_s/L\right)\right]\sin 2\alpha} \qquad \text{………(5.49)}$$

$$c = \frac{P_0 + \gamma H/2}{2[(\mathrm{FS} - \tan\phi/\tan\alpha)\sin(2\alpha)]^{-1} + r_s H/L} \qquad \text{………(5.50)}$$

Li and Aubertin (2012) during the field studies have observed the presence of tension crack at a depth of $H_t = \frac{2c}{\gamma\tan(45° - \phi/2)}$. So, considering the same, the equivalent wedge block width (B_e) that can fail can be determined using equation 5.51.

$$B_e = (H - H_t)/\tan\alpha \qquad \text{………(5.51)}$$

The above-mentioned relations (Eq. 5.46 to 5.51) are known as the modified Mitchell (MM) solution. It gives a proper estimation of the required strength of backfill as compared to the original Mitchell et al. (1982) solution. However, the MM solution also does not consider the normal and shear stresses along the fill-side wall interface, fill-back wall interface and the friction along the wall interface.

Numerical:

1. Unit weight of fill of 21.2 kN/m3, stope height 50 m, thickness of stope as 20 m, stope width along strike 25, fill friction angle as 32 degree, using Mitchell et al. (1982) solution determine the required strength of backfill, consider FS 1.
Answer: Cohesion: 123 kPa, UCS: 303 kPa

2. For the problem numerical in question 1 for FS 2 calculate the required strength.

Answer:

3. Unit weight of fill of 21.2 kN/m3, stope height 50 m, thickness of stope as 20 m, stope width along strike 25, fill friction angle as 32 degree, adherence ratio as 0.5, no overlying load, using Li & Aubertin solution determine the required strength of backfill, consider FS 1.
Answer: Cohesion: 117 kPa, UCS: 422 kPa

4. Unit weight of fill of 21.2 kN/m3, stope height 30 m, thickness of stope as 20 m, stope width along strike 25, fill friction angle as 32 degree, adherence ratio as 0.5, no overlying load, using Li & Aubertin solu tion determine the required strength of backfill, consider FS 1.
Answer:

ANNEXURE-6.1
Calculation of Backfill Slurry Parameters

$$\rho_b = Bulk\ Density,\ kg/m^3$$

$$S = Specific\ Gravity$$

$$C_v = Concentration\ by\ Volume$$

$$C_w = Concentration\ by\ Weight$$

1. $SLURRY\ DENSITY, kg/m^3 = \rho = 1 + C_V(S-1) = \frac{S}{S-C_w(S-1)}$

2. $POROSITY = P = \left[1 - \frac{\rho_b}{S}\right]$

3. $BULK\ DENSITY = \rho_b = (1-P).S$

4. *VOID RATIO* = $e = \frac{P}{(1-P)}$

5. $\left(\frac{WATER}{SOLID}\right)_{V/V} = \frac{(1-C_V)(1-P)}{C_V} = \frac{(1-C_V)\left(1-\frac{\rho_b}{S}\right)}{C_V} = \rho_b \times \left(\frac{WATER}{SOLID}\right)_{W/W}$

6. $\left(\frac{WATER}{SOLID}\right)_{w/w} = \left(\frac{1}{C_W} - 1\right)$

7. $C_V = \frac{C_w}{S-C_w\ (S-1)} = \frac{(1-P)}{\left(\frac{WATER}{SOLID}\right)_{V/V}+(1-P)} = \frac{1}{1+S\left(\frac{WATER}{SOLID}\right)_{w/w}} = \frac{\frac{\rho_b}{S}}{\left(\frac{WATER}{SOLID}\right)_{V/V}+\left(\frac{\rho_b}{S}\right)}$

8. $C_W = \frac{S.C_V}{1+C_V(S-1)} = \frac{1}{1+\left(\frac{WATER}{SOLID}\right)_{w/w}}$